

MAT 426 - Topics in Linear Algebra – Exam 1 Review Exercises

Below are some questions to help you prepare for Exam 1 on Tuesday, March 10. This list is not meant to be exhaustive and some topics may have been missed. Any topics that I have failed to include here are still considered reasonable exam material. Be sure to study all materials considered in class and in the homework. I have included selected answers as well - but these may contain mistakes as I have not double checked them.

1. Find the LU factorization of the following matrices.

(a) $A = \begin{bmatrix} 2 & 3 & 1 \\ 4 & 7 & 7 \\ 6 & 18 & 22 \end{bmatrix}$

(b) $A = \begin{bmatrix} 3 & 2 & 1 & 4 \\ 2 & 1 & 5 & 7 \\ 1 & 3 & 2 & 6 \\ 4 & 2 & 3 & 8 \end{bmatrix}$

(c) $A = \begin{bmatrix} 2 & 1 & 1 & 3 \\ 4 & 3 & 3 & 7 \\ 8 & 7 & 9 & 19 \end{bmatrix}$

(d) $A = \begin{bmatrix} 2 & 1 & 3 & 2 & 4 \\ 3 & 2 & 4 & 1 & 5 \\ 1 & 3 & 2 & 4 & 2 \\ 4 & 1 & 5 & 3 & 6 \\ 2 & 4 & 3 & 5 & 7 \end{bmatrix}$

2. Describe how to solve the system $A\mathbf{x} = \mathbf{b}$ using the LU factorization of the matrix A .
3. Suppose that A is nonsingular and has an LU factorization. Prove that this factorization is unique.
4.
 - (a) How many floating point operations are needed to solve an upper triangular system $U\mathbf{x} = \mathbf{y}$?
 - (b) How many floating point operations are needed to solve a unit lower triangular system $L\mathbf{y} = \mathbf{b}$?
5.
 - (a) How many floating point operations are needed to find the LU decomposition of a matrix A ?
 - (b) Performing row operations to row reduce $[A|I]$ to $[I|A^{-1}]$ is equivalent to solving n linear systems $A\mathbf{x} = \mathbf{e}_i$, where $\mathbf{e}_i$ is the i th column of the identity matrix I . How many floating point operations are needed to solve these n linear systems using the LU decomposition of A ?
 - (c) How many floating point operations are needed to find the inverse of A ?
 - (d) How many floating point operations are needed to multiply $A^{-1}\mathbf{b}$?
 - (e) How many floating point operations are needed to solve $A\mathbf{x} = \mathbf{b}$ by finding the inverse of A ?
6. State two reasons why the $PA = LU$ factorization is needed.

7. Find the $PA = LU$ factorization of the following matrices. Be comfortable answering either by giving P , L , and U or A and $\mathbf{r}$ with L and U both stored in A .

$$(a) \quad A = \begin{bmatrix} 0 & 2 & 1 \\ 1 & -2 & 0 \\ 3 & 1 & 4 \end{bmatrix}$$

$$(b) \quad A = \begin{bmatrix} 2 & 3 & 1 \\ 4 & 7 & 7 \\ 6 & 18 & 22 \end{bmatrix}$$

$$(c) \quad A = \begin{bmatrix} 0 & 2 & 1 & 1 \\ 1 & 1 & 0 & 2 \\ 2 & 4 & 1 & 3 \\ 1 & -1 & 2 & 1 \end{bmatrix}$$

$$(d) \quad A = \begin{bmatrix} 0 & 2 & 1 & 3 & 1 \\ 1 & 1 & 0 & 2 & 4 \\ 2 & 3 & 1 & 5 & 7 \\ 1 & -1 & 2 & 1 & 0 \end{bmatrix}$$

8. Gaussian elimination with partial pivoting directly yields the factorization $MA = U$ where M is a permuted elementary matrix and U is an upper triangular matrix. For a 4×4 matrix A , show how M is formed and how the $PA = LU$ factorization is extracted.
9. Outline the algorithm for computing the $PA = LU$ factorization of a matrix A .
10. Define the inner product of two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$.
11. Prove the following properties of inner products:
- $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$
 - $\mathbf{u} \cdot \mathbf{u} \geq 0$
 - $\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2 = 4\mathbf{u} \cdot \mathbf{v}$
12. Let $\mathbf{u} = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 0 \\ -5 \\ 4 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}$. Compute the following.
- $\mathbf{u} \cdot \mathbf{v}$
 - $\|\mathbf{u} - \mathbf{v}\|$
 - $\mathbf{u} \cdot \mathbf{v} + 2\|\mathbf{w}\|^2$
 - $\text{proj}_{\mathbf{w}} \mathbf{u}$
13. Prove the Triangle Inequality using the Cauchy-Schwarz inequality.
14. (a) Define the orthogonal complement $W^\perp$ of a subspace W of $\mathbb{R}^n$.
 (b) Prove that $W^\perp$ is a subspace of $\mathbb{R}^n$.
15. Determine which of the following sets are orthogonal. If the set is orthogonal, use it to form an orthonormal set of vectors.

$$(a) \quad S = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$(b) S = \left\{ \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix} \right\}$$

$$(c) S = \left\{ \begin{bmatrix} 1 \\ 2 \\ -1 \\ 0 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 1 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ -1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -3 \\ 2 \\ 2 \\ -2 \end{bmatrix} \right\}$$

16. Prove the following: If $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p\}$ is an orthogonal set of nonzero vectors in $\mathbb{R}^n$, then S is linearly independent.
17. Prove the following: Let $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$ and let $\mathbf{y}$ be any vector in W . Then, the weights $c_1, c_2, \dots, c_p$ in the linear combination

$$\mathbf{y} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \dots + c_p\mathbf{u}_p$$

are given by

$$c_j = \frac{\mathbf{y} \cdot \mathbf{u}_j}{\mathbf{u}_j \cdot \mathbf{u}_j}$$

18. Find the orthogonal projection of $\mathbf{y} = \begin{bmatrix} 2 \\ -3 \\ 7 \end{bmatrix}$ onto $\mathbf{u} = \begin{bmatrix} -3 \\ 1 \\ 4 \end{bmatrix}$. Then, write $\mathbf{y}$ as the sum of two vectors - one in $\text{Span}(\mathbf{u})$ and one orthogonal to $\mathbf{u}$.
19. For each of the following, show that S is an orthogonal set and then express $\mathbf{y}$ as a linear combination of the elements in S .

$$(a) S = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \right\}, \mathbf{y} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$$

$$(b) S = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \\ 3 \end{bmatrix} \right\}, \mathbf{y} = \begin{bmatrix} 5 \\ 7 \\ 11 \\ 9 \end{bmatrix}$$

$$(c) S = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} -4 \\ 3 \\ 3 \\ 1 \end{bmatrix} \right\}, \mathbf{y} = \begin{bmatrix} \frac{1}{6} \\ \frac{1}{3} \\ \frac{3}{2} \\ 1 \end{bmatrix}$$

$$(d) S = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 3 \\ -1 \\ 3 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\}, \mathbf{y} = \begin{bmatrix} 0 \\ 1 \\ 3 \\ 1 \\ 3 \end{bmatrix}$$

20. Show that the matrix $A = \begin{bmatrix} \frac{1}{\sqrt{3}} & -\frac{1}{2\sqrt{6}} & \frac{1}{2\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{2\sqrt{6}}{1} & -\frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{6}} & \frac{1}{2\sqrt{2}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{2\sqrt{6}} & \frac{1}{2\sqrt{2}} \\ 0 & \frac{1}{2\sqrt{6}} & \frac{1}{2\sqrt{2}} \end{bmatrix}$ is orthonormal.

21. Show that the matrix $A = \begin{bmatrix} \cos(\alpha) \sin(\beta) & -\sin(\alpha) & -\cos(\alpha) \sin(\beta) \\ \sin(\alpha) \cos(\beta) & \cos(\alpha) & -\sin(\alpha) \sin(\beta) \\ \sin(\beta) & 0 & \cos(\beta) \end{bmatrix}$ is orthogonal.
22. Prove the following: Let U be an $n \times n$ matrix. The following statements are equivalent:
- (a) U is orthogonal
 - (b) $\|U\mathbf{x}\| = \|\mathbf{x}\|$ for every $\mathbf{x}$ in $\mathbb{R}^n$.
 - (c) $(U\mathbf{x}) \cdot (U\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$ for every $\mathbf{x}, \mathbf{y}$ in $\mathbb{R}^n$

Selected Answers

1.

$$(a) \quad L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 9 & 1 \end{bmatrix}, U = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 1 & 5 \\ 0 & 0 & -26 \end{bmatrix}$$

$$(b) \quad L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{2}{3} & 1 & 0 & 0 \\ \frac{1}{3} & -7 & 1 & 0 \\ \frac{4}{3} & 2 & -\frac{7}{32} & 1 \end{bmatrix}, U = \begin{bmatrix} 3 & 2 & 1 & 4 \\ 0 & -\frac{1}{3} & \frac{13}{3} & \frac{13}{3} \\ 0 & 0 & 32 & 35 \\ 0 & 0 & 0 & \frac{53}{32} \end{bmatrix}$$

$$(c) \quad L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{bmatrix}, U = \begin{bmatrix} 2 & 1 & 1 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 4 \end{bmatrix}$$

$$(d) \quad L = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ \frac{3}{2} & 1 & 0 & 0 & 0 \\ \frac{1}{2} & 5 & 1 & 0 & 0 \\ 2 & -2 & \frac{2}{3} & 1 & 0 \\ 1 & 6 & 1 & \frac{6}{11} & 1 \end{bmatrix}, U = \begin{bmatrix} 2 & 1 & 3 & 2 & 4 \\ 0 & \frac{1}{2} & -\frac{1}{2} & -2 & -1 \\ 0 & 0 & 3 & 13 & 5 \\ 0 & 0 & 0 & \frac{11}{3} & -\frac{2}{3} \\ 0 & 0 & 0 & 0 & \frac{48}{11} \end{bmatrix}$$

4. (a) n^2

(b) $n(n-1)$

5. (a) $\frac{2}{3}n^3 + \mathcal{O}(n^2)$

(b) $2n^3$

(c) $\frac{8}{3}n^3 + \mathcal{O}(n^2)$

(d) $2n^2 - n$

(e) $\frac{8}{3}n^3 + \mathcal{O}(n^2)$

6. The LU factorization algorithm can fail because it encounters a zero pivot and the LU factorization algorithm can provide an incorrect factorization because it encountered a pivot close to zero resulting a very large multiplier.

7.

$$(a) \quad P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, L = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{3} & 1 & 0 \\ 0 & -\frac{6}{7} & 1 \end{bmatrix}, U = \begin{bmatrix} 3 & 1 & 4 \\ 0 & -\frac{7}{3} & -\frac{4}{3} \\ 0 & 0 & -\frac{1}{7} \end{bmatrix}$$

$$(b) \quad P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 9 & 1 \end{bmatrix}, U = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 1 & 5 \\ 0 & 0 & -26 \end{bmatrix}$$

$$(c) \quad P = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{2} & 1 & 0 & 0 \\ 0 & -\frac{2}{3} & 1 & 0 \\ \frac{1}{2} & \frac{1}{3} & -\frac{1}{5} & 1 \end{bmatrix}, U = \begin{bmatrix} 2 & 4 & 1 & 3 \\ 0 & -3 & \frac{3}{2} & -\frac{1}{2} \\ 0 & 0 & 2 & \frac{4}{3} \\ 0 & 0 & 0 & \frac{7}{15} \end{bmatrix}$$

$$(d) \quad P = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{2} & 1 & 0 & 0 \\ 0 & -\frac{4}{5} & 1 & 0 \\ \frac{1}{2} & -\frac{1}{5} & -\frac{4}{11} & 1 \end{bmatrix}, U = \begin{bmatrix} 2 & 3 & 1 & 5 & 7 \\ 0 & -\frac{5}{2} & \frac{3}{2} & -\frac{3}{2} & -\frac{7}{2} \\ 0 & 0 & \frac{11}{5} & \frac{9}{5} & -\frac{9}{5} \\ 0 & 0 & 0 & \frac{5}{11} & \frac{6}{11} \\ 0 & 0 & 0 & \frac{5}{11} & \frac{6}{11} \end{bmatrix}$$

7.

$$\begin{aligned} U &= M_3 P_3 M_2 P_2 M_1 P_1 A \\ &= M_3 (P_3 M_2 P_3) (P_3 P_2 M_1 P_2 P_3) (P_3 P_2 P_1) A \\ &= M'_2 M'_2 M'_1 A \end{aligned}$$

where $M'_3 = M_3$, $M'_2 = P_3 M_2 P_3$, and $M'_1 = P_3 P_2 M_1 P_2 P_3$. Then, $P = P_3 P_2 P_1$ and $L = (M'_1)^{-1} (M'_2)^{-1} (M'_3)^{-1}$. (Why are $M_k'^{-1}$ lower triangular? Why are the M_k elementary matrices?)

8. LU FACTORIZATION USING GEPP.

Input: An $n \times n$ matrix A

Outputs:

1. Matrix A contains the upper triangular matrix U in the upper triangle of A and the unit lower triangular matrix L in the lower triangle of A
2. An $n - 1$ vector containing the row swaps

For $k = 1, 2, \dots, n - 1$ do

- (a) (Find the pivot row). Find r_k such that $|a_{r_k, k}| = \max_{k \leq i \leq n} |a_{i, k}|$. If $a_{r_k, k} = 0$, stop. Otherwise, continue.
- (b) (Interchange rows r_k and k). $a_{kj} \longleftrightarrow a_{r_k, j}$ $j = 1, \dots, n$.
- (c) (Form the multipliers) $a_{i, k} \equiv m_{i, k} = \frac{a_{i, k}}{a_{k, k}}$ ($i = k + 1, \dots, n$)
- (d) (Update the entries) $a_{i, j} \equiv a_{i, j} - m_{i, k} a_{k, j} = a_{i, j} - a_{i, k} a_{k, j}$ $i = k + 1, \dots, n; j = k + 1, \dots, n$

12. (a) -23

(b) 10

(c) -9

(d) $\text{proj}_{\mathbf{w}} \mathbf{u} = \begin{bmatrix} 11/7 \\ 11/14 \\ -33/14 \end{bmatrix}$

15. (a) $S' = \left\{ \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{bmatrix}, \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix} \right\}$

(b) Not orthogonal

$$(c) \ S' = \left\{ \begin{bmatrix} \frac{1}{\sqrt{15}} \\ \frac{2}{\sqrt{15}} \\ -\frac{1}{\sqrt{15}} \\ 0 \\ \frac{3}{\sqrt{15}} \end{bmatrix}, \begin{bmatrix} \frac{2}{\sqrt{15}} \\ -\frac{1}{\sqrt{15}} \\ \frac{1}{\sqrt{15}} \\ \frac{3}{\sqrt{15}} \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ \frac{1}{\sqrt{7}} \\ -\frac{1}{\sqrt{7}} \\ \frac{2}{\sqrt{7}} \\ \frac{1}{\sqrt{7}} \end{bmatrix}, \begin{bmatrix} \frac{3}{\sqrt{12}} \\ 0 \\ -\frac{1}{\sqrt{12}} \\ \frac{1}{\sqrt{12}} \\ -\frac{1}{\sqrt{12}} \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{22}} \\ -\frac{3}{\sqrt{22}} \\ \frac{2}{\sqrt{22}} \\ \frac{2}{\sqrt{22}} \\ -\frac{2}{\sqrt{22}} \end{bmatrix} \right\}$$

$$18. \ \mathbf{y} = \begin{bmatrix} -\frac{57}{26} \\ \frac{19}{26} \\ \frac{38}{53} \\ \frac{13}{13} \end{bmatrix} + \begin{bmatrix} \frac{109}{26} \\ \frac{97}{26} \\ \frac{53}{13} \end{bmatrix}.$$

$$19. \quad (a) \ \mathbf{y} = \frac{4}{3}\mathbf{u}_1 + \frac{3}{2}\mathbf{u}_2 - \frac{5}{6}\mathbf{u}_3$$

$$(b) \ \mathbf{y} = 6\mathbf{u}_1 + 4\mathbf{u}_2 + 3\mathbf{u}_3$$

$$(c) \ \mathbf{y} = \frac{1}{2}\mathbf{u}_1 + \frac{1}{3}\mathbf{u}_2 + \frac{1}{6}\mathbf{u}_3$$

$$(d) \ \mathbf{y} = \frac{2}{3}\mathbf{u}_1 + \frac{2}{3}\mathbf{u}_2 + \mathbf{u}_3$$